

1(a)	angle (subtended at centre of circle) when arc length = radius	B1
1(b)	$\omega = 2\pi / T$	C1
	$= 2\pi / (1.0 \times 60 \times 60)$ $= 1.7 \times 10^{-3} \text{ rad s}^{-1}$	A1
1(c)(i)	angle $= 1.7 \times 10^{-3} \times 1400$ $= 2.4 \text{ rad}$	A1
1(c)(ii)	$L = \text{arc length} / \text{angle}$ $= 0.44 / 2.4$ or $L = 0.44 \times (3600 / 1400) / 2\pi$	C1
	$L = 0.18 \text{ m}$	A1
1(c)(iii)	$a = r\omega^2$	C1
	$= 0.18 \times (1.745 \times 10^{-3})^2$ $= 5.5 \times 10^{-7} \text{ m s}^{-2}$	A1
1(d)	centripetal acceleration is negligible compared with acceleration of free fall or numerical comparison establishing answer to (c)(iii) $\ll 9.81$	B1
	resultant force is negligible compared with weight (of modelling clay) (so variation is negligible) or force exerted by minute hand (approximately) equal (and opposite) to weight of modelling clay	B1